

TASVIRLARNI TAHLIL QILISH VA ANGLASH · Amaliy mashg'ulot (Practice)

8-amaliy mashg'ulot. Obyektlarni tanib olishning klassik usullari

2026–2027 o'quv yili

Mavzu. Shablon moslashtirish (SSD, NCC), HOG deskriptori, chiziqli SVM, k-NN, PCA (Eigenfaces), BoVW, kichik CNN; sliding window, tasvir piramidasi, IoU, NMS, soft-NMS; precision, recall, F1-score; integral tasvir, Haar belgilari, AdaBoost, kaskad, hard negative mining.

Maqsadlar. Talaba: (1) NCC, HOG ovozi va L2-Hys, SVM margin'i, IoU, NMS, integral tasvir va AdaBoost vaznlarini qo'lda hisoblaydi; (2) NCC va HOG invariantligini, PCA dagi $N \ll D$ hiylasini isbotlaydi; (3) `scikit-image`, `scikit-learn`, OpenCV va PyTorch yordamida tasniflagich va suzuvchi oyna detektorini quradi; (4) detektorni precision, recall, F1-score bilan halol baholaydi.

Baholash (100 ball). 1–17-topshiriqlar: har biri 2 ball (34); 18–30-topshiriqlar: har biri 4 ball (52); kod sifati, takrorlanuvchanlik (seed = 42) va xulosalar: 14 ball. Javoblar kamida 4 xona aniqlikda yoki topshiriqda ko'rsatilgan formatda beriladi.

Muhit (setup)

Python 3.11 yoki 3.12, faqat CPU: `python -m venv .venv; pip install -r requirements.txt` (numpy==2.4.4, scipy==1.17.1, pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0, opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0). Ma'lumotlar oflayn: `skimage.data.coins()`, `skimage.data.camera()`, `sklearn.datasets.load_digits()`, `make_blobs` va sintetik sahnalar. Bitta `solutions.py` (har topshiriqqa funksiya va `main`), natijalar `./out/` ga; `python -W error::DeprecationWarning solutions.py` xatosiz ishlashi shart.

Umumiy yordamchi kod (22–28, 30-topshiriqlar):

```
import numpy as np, cv2
from skimage import data
from skimage.feature import hog
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
SEED = 42
COIN = (144, 236, 201, 288) # coins dagi shablon tanga (x1, y1, x2, y2)
def digits_split(): # 1797 ta 8x8 raqam (0..16) -> 1257 train / 540 test
    d = load_digits()
    return train_test_split(d.images, d.target, test_size=0.3,
                           stratify=d.target, random_state=SEED)
def hog_digit(img): # 8x8 -> 324 o'lchovli HOG
    return hog(img, orientations=9, pixels_per_cell=(2, 2),
               cells_per_block=(2, 2), block_norm="L2-Hys")
def iou(a, b): # a, b = (x1, y1, x2, y2)
    w = max(0, min(a[2], b[2]) - max(a[0], b[0]))
    h = max(0, min(a[3], b[3]) - max(a[1], b[1]))
    I = w * h
    return I / ((a[2]-a[0])*(a[3]-a[1]) + (b[2]-b[0])*(b[3]-b[1]) - I)
```

1. Qo'lda hisoblash, hosil qilish va tushunchalar

- NCC va SSD.** 1D shablon $T = (1, 2, 3, 4)$, oyna $W = (2, 5, 6, 9)$: NCC va SSD ni hisoblang. $3W + 7$ va $-W$ uchun NCC nimaga teng?
- Nega normallashtirish?** a) $W \mapsto aW + b$ da NCC $a > 0$ bo'lsa o'zgarmasligini, $a < 0$ da ishorasi almashishini isbotlang. b) $T = (1, 0, 1)$, $W_1 = (1, 0, 1)$, $W_2 = (9, 9, 8)$: $\langle W, T \rangle$, NCC va SSD ni hisoblang. Normallanmagan korrelyatsiya qaysi oynani tanlaydi va nega bu xato?
- HOG o'lchami.** 48×96 oyna (kenglik $\times$ balandlik), 8×8 katak, 2×2 blok, qadam 1 katak, $B = 9$: D ni toping. 8×8 raqam, 2×2 katak, 2×2 blok uchun-chi?
- HOG ovozi.** Pikselda $g_x = -20$, $g_y = 20$. Ishorasiz yo'nalish θ va m ni toping; markazlari $10^\circ, 30^\circ, \dots, 170^\circ$ bo'lgan kataklar orasida chiziqli taqsimlangan (Dalal–Triggs) ovozlarni hisoblang.
- L2-Hys.** Blok vektori $v = (1, 1, 2, 4, 10)$ uchun L2 normallashtirish, 0,2 da kesish va qayta L2 normallashtirish natijasini toping ($\varepsilon = 0$). Kesish nima beradi?

6. **HOG invariantligi (isbot).** $I \mapsto aI + b$ ($a > 0$, kesilishsiz) da gradient yo'nalishi o'zgarmasligini va $\varepsilon \rightarrow 0$ da HOG deskriptori o'zgarmasligini isbotlang. $I \mapsto \sqrt{I}$ da nega bu tasdiq bajarilmaydi?
7. **SVM qo'lda.** Manfiy: (1, 1), (2, 3); musbat: (4, 2), (5, 4). Qattiq margin'li SVM uchun w , b , margin $2/\|w\|$ ni toping. Qaysi nuqtalar $f = \pm 1$ chiziqlarida yotadi? (3, 3) qaysi sinfga tegishli? (*Ko'rsatma:* eng yaqin qarama-qarshi juftlik (2, 3)–(4, 2).)
8. **Hinge loss.** $w = (1, -1)$, $b = 0,5$, $C = 1$. Namunalar: (2, 1) va (1, 1) — $y = +1$; (0, 1) va (1, 0) — $y = -1$. Har biri uchun $yf(x)$, hinge loss va maqsad funksiya $\frac{1}{2}\|w\|^2 + C \sum \xi_i$ ni toping. Qaysi nuqta noto'g'ri tasniflangan, qaysilari margin ichida?
9. **k-NN va masshtab.** O'qitish: (1, 200) va (1,5; 270) — sinf 0; (4, 205) va (4,5; 260) — sinf 1. So'rov $q = (1,2; 240)$. Evklid masofasi bilan 1-NN va 3-NN javobi; ikkinchi belgi 100 ga bo'linganda-chi? Xulosa.
10. **PCA qo'lda.** Nuqtalar (2, 1), (4, 3), (0, 1), (2, 3). $\bar{x}$, $\Sigma = \frac{1}{N} X_c^T X_c$, xos qiymatlar, u_1 , birinchi komponenta tushuntirgan dispersiya ulushi va (4, 3) ning z_1 proyeksiyasini toping.
11. $N \ll D$ **hiylasi.** $X_c X_c^T v = \lambda v$, $\lambda > 0$ bo'lsa, $u = X_c^T v$ vektor $X_c^T X_c$ ning xos vektori ekanini isbotlang. 64×64 tasvirlar ($D = 4096$), $N = 400$ uchun ikki matritsaning o'lchami va `float64` dagi xotirasi.
12. **BoVW.** Lug'at (1D) {0, 5, 10}, tasvir deskriptorlari {1, 4, 6, 9, 10, 2}. Normallangan gistogramma h va $g = (0,5; 0,25; 0,25)$ bilan histogram intersection $K(h, g)$.
13. **Oynalar soni.** 320×240 tasvir, 64×64 oyna, qadam 8: bitta masshtabda nechta oyna? Piramida koeffitsiyenti $q = \sqrt{2}$ va $q = 1,25$ bo'lganda umumiy son nechta barobar oshadi (yuqori chegara)?
14. **IoU va NMS.** Qutilar (x_1, y_1, x_2, y_2) va ballar: $A = (0, 0, 40, 40)$ 0,95; $B = (5, 5, 45, 45)$ 0,9; $C = (50, 10, 90, 50)$ 0,8; $D = (20, 0, 60, 40)$ 0,7; $E = (52, 12, 88, 48)$ 0,6. IoU(A, B), (A, D), (B, D), (C, E), (C, D) ni hisoblang va greedy NMS natijasini $\rho = 0,5$ va $\rho = 0,3$ uchun toping.
15. **Detektorni baholash.** Etalonlar $G_1 = (2, 2, 42, 42)$, $G_2 = (48, 8, 88, 48)$. 14-topshiriqdagi detektsiyalar uchun (ball bo'yicha tartibda, IoU $\geq 0,5$, har etalon bir marta) TP, FP, precision, recall, F1 ni: (a) NMS siz; (b) $\rho = 0,5$; (c) $\rho = 0,3$ holatlarda toping.
16. **Integral tasvir va Haar.** $I = \begin{pmatrix} 5 & 1 & 2 & 1 \\ 3 & 4 & 0 & 2 \\ 0 & 2 & 1 & 1 \\ 2 & 1 & 3 & 0 \end{pmatrix}$. $S(r, c) = \sum_{r' \leq r, c' \leq c} I$ ni tuzing; 1..2-qatorlar va 1..3-ustunlar (0 dan sanaganda) yig'indisini 4 ta murojaat bilan toping; “chap yarim — o'ng yarim” va “yuqori yarim — pastki yarim” Haar belgilarini hisoblang.
17. **AdaBoost va kaskad.** a) 10 namuna, $w_i = 0,1$, h_1 3 tasida xato: ε_1 , α_1 , Z_1 , yangilangan vaznlar; xato namunalar umumiy vazni $1/2$ ekanini tekshiring. b) Kaskad: narxlar (0,01; 0,1; 1), o'tish ulushlari $p_1 = 0,25$, $p_2 = 0,2$; $\mathbb{E}[c]$ va tezlanish. c) 10 bosqich, $d_i = 0,998$, $f_i = 0,4$: D va F .

2. Dasturlash topshiriqlari

Har bir topshiriq — `solutions.py` dagi alohida funksiya (`task18...()`, ..., `task30...()`). Sonlarni ko'rsatilgan formatda chop eting.

18. **NCC qo'lda.** `ncc_map(S, T)` (“valid” xarita): surat — `np.einsum` va `sliding_window_view` bilan markazlashtirilgan shablon korrelyatsiyasi, maxraj — integral tasvir orqali lokal $\sum W$, $\sum W^2$. `I = data.coins()`, `T = I[236:288, 144:201]`. `cv2.matchTemplate(I, T, cv2.TM_CCORR_NORMED)` bilan `max|farq| (%.2e)`, `argmax(x, y)` va bir-biridan 20 pikseldan uzoq eng yaxshi 5 ta joy (x, y, NCC).
19. **Shablon moslashtirish (coins).** `find(S, T)` — `TM_CCORR_NORMED` maksimumi va quti. Sharoitlar: asl; `0.55I+40`; `gamma np.round(255*(I/255)*0.5)`; tasvir `cv2.resize(I, None, fx=1.25, fy=1.25)` (to'g'ri quti — `COIN × 1,25`, yaxlitlangan). Har biri uchun NCC (%.4f), quti va IoU (%.3f). So'ng masshtablangan tasvirda shablonni $s \in \{0,8; 0,9; \dots; 1,4\}$ ga o'zgartirib qidiring; eng yaxshi s ni tanlang. Yuqori NCC har doim to'g'ri javobmi?
20. **HOG katak gistogrammasi.** `hog_cell_manual(cell, nbins=9)`: $[-1, 0, 1]$ gradient, $\theta = \text{atan2}(g_r, g_c) \bmod 180^\circ$, qattiq (interpolyatsiyasiz) kataklarga ovoz, katak yuzasiga bo'lish (`scikit-image` uslubi); `cell` — 1 piksellik chegarasi bilan. Tekshiruv: `img = rng.integers(0, 256, (18, 18)) (default_rng(42))`, `hog(img, 9, (8, 8), (1, 1), block_norm="L1", feature_vector=False)` ning (1,1) katagi bilan `max|farq|`. Vertikal chegara (qatorlari (10,10,10,50,50,50)) va $I = 10(r+c)$ rampa uchun ovozlari. Natijani ma'ruzadagi interpolyatsiyali ovoz bilan solishtiring.
21. **HOG o'lchami va chidamliligi.** 32×32 , 48×96 , 64×128 oynalar uchun formula va `hog(np.zeros((h, w)), 9, (8, 8), (2, 2)).size`. `COIN` yamog'ini `cv2.resize(..., (32, 32), interpolation=cv2.INTER_AREA)` qilib, `0.55p+40`, `2p`, `16*sqrt(p)`, `p + N(0, 2) (default_rng(42))` uchun `max|dHOG|` va kosinus o'xshashlik.
22. **HOG + LinearSVC (load_digits).** Belgilar: xom piksel (64) va `hog_digit (324)`. `make_pipeline(StandardScaler(), LinearSVC(C=0.1, max_iter=20000, random_state=42))`. Test: asl va yoritishi o'zgargan `0.5*X+4`. Accuracy va macro-F1 (%.4f); HOG uchun eng ko'p uchragan 5 ta xato (soni, haqiqiy, bashorat).
23. **SVM va C.** 7-topshiriqni `SVC(kernel="linear", C=1e6)` bilan tekshiring (`coef_`, `intercept_`, `support_`, `dual_coef_`). `make_blobs(n_samples=200, centers=2, cluster_std=2.0, random_state=42)` da $C \in$

- $\{0,001;0,01;0,1;1;100\}$ uchun $\|w\|$, margin, tayanch vektorlar soni, train accuracy.
24. **k-NN.** Train qismida `StratifiedKFold(5, shuffle=True, random_state=42)` bilan $k \in \{1, 3, 5, 7, 9\}$ uchun CV accuracy (xom piksel va HOG); eng yaxshi k bilan test accuracy.
 25. **PCA (eigen-digits).** `PCA(n_components=r, random_state=42)`, $r \in \{5, 10, 20, 40\}$: tushuntirilgan dispersiya, PCA fazosida 1-NN test accuracy, test rekonstruksiya MSE. Oʻrtacha va 8 ta komponentani `out/m08_eigendigits.png` ga saqlang. $N \ll D$ hiylasi: `data.camera()` dan `rng.integers(0, 448, (20, 2))` burchakli 20 ta 64×64 boʻlak; $X_c X_c^T$ orqali topilgan 3 ta xos vektor uchun nisbiy qoldiq $\|X_c^T X_c u - \lambda u\|/\lambda$.
 26. **BoVW.** Har raqamdan 4×4 yamoqlar, qadam 2 (9 ta, 16 oʻlchov; `sliding_window_view(A, (4, 4), axis=(1, 2))[:, ::2, ::2]`). Lugʻat: `KMeans(V, n_init=4, random_state=42)`, $V \in \{8, 16, 32\}$. Gistogramma (/9) + `SVC(kernel="precomputed", C=10)` histogram intersection yadrosi bilan; taqqoslash uchun joylashuvli variant (har yamoq pozitsiyasi uchun one-hot soʻz, 9V oʻlchov) + `LinearSVC(C=0.1)`. Test accuracy.
 27. **Kichik CNN.** `Conv2d(1,16,3,padding=1)-ReLU-MaxPool2d(2)-Conv2d(16,32,3,padding=1)-ReLU-MaxPool2d(2)-Flatten-Linear(128,10)`. Kirish $X/16$; `torch.manual_seed(42)`, `torch.use_deterministic_algorithms(True)`, `torch.set_num_threads(1)`; Adam, `lr=1e-2`, batch 64 (`torch.randperm` seed 42 li Generator bilan), 30 epoxa. Parametrlar soni; 1, 5, 10, 20, 30-epoxalarda train loss va test accuracy; 0.5×4 test. HOG + SVM bilan solishtiring.
 28. **Sliding window + NMS (sintetik sahna).** 128×128 fon `clip(N(0,1), 0)`; unga test toʻplamidan 10 ta raqam (4 tasi "0") `np.kron(img, np.ones((2, 2)))` (16×16) qilib kesishmasiz joylashtiriladi (`default_rng(43)`, 3 ta sahna). Detektor: HOG (`hog(p, 9, (4, 4), (2, 2))`, 324) + `LinearSVC(C=0.1, max_iter=50000)`; musbat — train "0" lari va ± 1 px siljishlar; manfiy — boshqa raqamlar, ± 4 px siljigan "0", X yamoq (`default_rng(42)`). Oyna 16×16 , qadam 2, ball > 0 ; NMS $\rho = 0,3$. NMS gacha va keyin: qutilar, TP, precision, recall, F1. Soʻng train raqamlaridan 5 ta sahnada (`default_rng(44)`) hard negative mining (IoU $< 0,3$ boʻlgan topilmalar) va qayta oʻqitish. `out/m08_detect.png`.
 29. **IoU/NMS kutubxonalar bilan.** `default_rng(42)`: 200 ta quti ($x_1, y_1 \sim U(0, 200)$, $w, h \sim U(20, 60)$, 0,1 gacha yaxlitlangan), ballar $U(0, 1)$ (4 xona). Oʻz nms ingizni `torchvision.ops.nms` va `cv2.dnn.NMSBoxes` (`(x,y,w,h)` format!) bilan $\rho \in \{0,3;0,5\}$ da solishtiring; IoU matritsasini `torchvision.ops.box_iou` bilan. 14-topshiriq qutilariga Gauss soft-NMS ($\sigma = 0,5$) qoʻllab, yakuniy ballarni chiqaring.
 30. **Integral tasvir, Haar, AdaBoost, kaskad.** `integral_image(I)` ($((H + 1) \times (W + 1))$ va `rect_sum`; `cv2.integral(data.camera())` bilan tenglik va 1000 ta tasodifiy toʻrtburchakda tekshiruv. 8×8 raqamlar uchun barcha ikki toʻrtburchakli Haar belgilari (chap-oʻng: $h = 1..8$, $w \in \{2, 4, 6, 8\}$; yuqori-past: simmetrik), "0" ga qarshi qolganlar; `AdaBoostClassifier(estimator=DecisionTreeClassifier(max_depth=1), n_estimators=100, random_state=42)`: accuracy, precision, recall, F1. Kaskad: 1-bosqich — dastlabki 5 ta stump (`staged_decision_function`), ostona — train musbatlari balli 1%-kvantili; 2-bosqich — toʻliq model. $p_1, \mathbb{E}[c] = 0,05 + p_1$, tezlanish, kaskad precision/recall/F1.

Hujjatlar

https://docs.opencv.org/4.x/df/dfb/group__imgproc__object.html (matchTemplate); https://docs.opencv.org/4.x/d4/dc6/tutorial_py_template_matching.html; https://docs.opencv.org/4.x/d6/d0f/group__dnn.html (NMSBoxes); <https://scikit-image.org/docs/stable/api/skimage.feature.html> (hog); <https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html>; https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html; <https://docs.pytorch.org/vision/stable/generated/torchvision.ops.nms.html>; <https://docs.pytorch.org/docs/stable/nn.html>.